

Munkres Review

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1 Introduction

If $L = [l, u]$ is linear order with the least upper bound property, then L is compact. Suppose $\mathcal{A}$ is a covering of L . Take the elements of $\mathcal{A}$ to be basis elements of the order topology, i.e. of the form $[l, a), (a, b), (b, u]$.

Let $m = \sup\{b \in L : [l, b] \text{ is finitely covered}\}$ (well-defined since $[l, l]$ is finitely covered by some element in $\mathcal{A}$ and that L is bounded above). If $m = u$, pick basis $(c, u] \in \mathcal{A}$ of u . There exists $c < k \leq u$ such that $[l, k]$ is finitely covered, so $[l, u]$ is finitely covered.

Else, $m < u$. Pick a basis $m \in (c, d) \in \mathcal{A}$. There exists $c < k \leq m < d$ such that $[l, k]$ is finitely covered (else, contradicts supremum). Then $[l, d]$ is finitely covered. If $(m, d) = \emptyset$, then take any element in $\mathcal{A}$ covering $\{d\}$, so $[l, d]$ is finitely covered, cont. sup. m . Else, there exists $m < e < d$, so $[l, e]$ is finitely covered, cont. sup. again.

If X is first-countable {limit-point compact} Hausdorff, then X is sequentially compact.

Suppose $\{x_n\} = (x_1, x_2, x_3, \dots)$ is a sequence. If for some i , there are infinitely many terms equal to x_i , then we have a convergent subsequence. Otherwise, there are finitely many instances of each x_i . Since $\{x_n\}$ is infinite, there are infinitely many distinct x_i . Hence, assume $\{x_n\}$ is a sequence of distinct x_i .

Suppose $\{x_n\}$ has a limit point x_∞ . Assume $x_\infty \notin \vec{x}$. Let $\{U_n\}$ be a countable neighborhood basis of x_∞ . Construct $\{a_n\}$ inductively as follows: assume we have $x_{a_1}, x_{a_2}, \dots, x_{a_k}$. Since X is Hausdorff, there exists a neighborhood W_{k+1} disjoint to $x_1, x_2, x_3, \dots, x_{a_k}$. Then $W_{k+1} \cap U_{k+1}$ is a neighborhood of x_∞ intersecting $\{x_n\}$ at indices after a_k . Pick $x_{a_{k+1}}$ to be any such term.

Now if $U(x_\infty)$ is any open set, there exists $U_n(x_\infty) \subset U(x_\infty)$. Since $x_{a_n} \in U_n$, $U \cap \{x_{a_n}\} \neq \emptyset$, so $\{x_{a_n}\}$ indeed converges to x_∞ .

Conversely, if X is sequentially compact, then X is limit-point compact. To

see this, let $A \subset X$ be infinite. Pick $\{x_n\} \subset A$ to have infinitely many distinct elements. Pick $\{x_{a_n}\} \subset \{x_n\}$ to converge to x_∞ . Then x_∞ is a limit point of $\{x_{a_n}\}$ and therefore A .

If $\alpha \in \omega_1$, there exist $\beta, \gamma \in \text{lOrd} \cap \omega_1$ st $\beta \leq \alpha < \gamma$ and if $\beta < \delta < \gamma$, then $\delta \notin \text{lOrd}$. Let $\gamma = \alpha + \omega \in \text{lOrd} \cap \omega_1$. Let $\beta = \sup\{\alpha \cap \text{lOrd}\} \leq \sup \alpha \leq \alpha$. Note $\beta \in \text{lOrd} \cap \omega_1$. For $\delta \in \text{lOrd}$, if $\beta < \delta < \alpha$, then $\delta > \beta = \sup\{\text{lOrd} \cap \alpha\}$. If $\alpha \leq \delta < \gamma$, let $\delta = \alpha + \epsilon$ for some nonzero $\epsilon \in \text{lOrd}$ ($\epsilon \neq 0$, the case for $\alpha \in \text{lOrd}$ has $\beta = \alpha$), but $\epsilon < \omega$, cont. If $\alpha \notin L$, pick $\beta < \alpha < \gamma$ as defined above, so L is closed.

ω_1 is completely normal, not Lindelof (thus not compact), locally metrizable but not metrizable, doesn't have countable dense subset, uncountable basis.